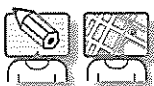


Name:

Group	Common properties	Colour key used on periodic table template	Starting element → finishing element
V	The first two members of this group are non-metals, while the next two members (arsenic and antimony) are metalloids. The final member of the group is a metal.		
VI	The first two elements in this group are non-metals and can form -2 ions. The next two members, selenium and tellurium, are metalloids.		
VII	This group is called the 'halogens' (salt producers). They are all strong electron acceptors and form diatomic molecules. The last member of the group is a metalloid.		
VIII	This group is known as the noble gases. The atoms of noble gases are very stable and unreactive.		

Extra for experts

- 6 Research the special properties of the transition metals, lanthanides and actinides.



Figuring out electron configurations—page 1

Name: _____

Skills: applying, interpreting

Electrons are arranged in shells or orbits around the nucleus. Some filling guidelines for electrons within these shells are summarised below:

- The first shell or orbit is called the 'K' shell and is identified as principal quantum number 1. It can hold a maximum of two electrons.
- The second shell or orbit is called the 'L' shell and is identified as principal quantum number 2. It can hold a maximum of eight electrons.
- The third shell or orbit is called the 'M' shell and is identified as principal quantum number 3. It can hold a maximum of 18 electrons.

Electron configurations can be represented by showing the number of electrons in each of the principal quantum numbers. They can also be shown visually by electron shell model diagrams. For example, carbon, which has an atomic number of 6, has six electrons to be placed in the electron cloud. This is shown in Figure 1.

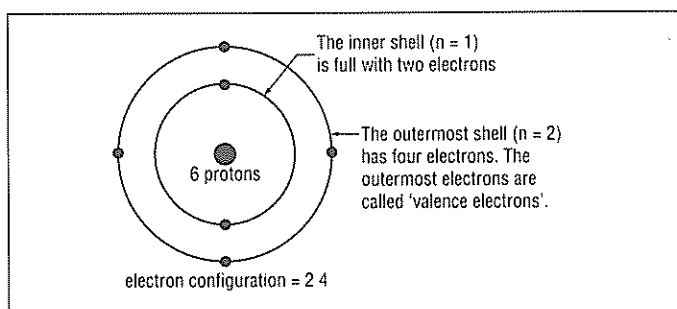


Fig 1 Electron configuration of carbon as an electron shell model

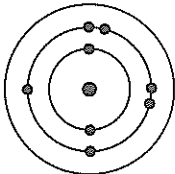
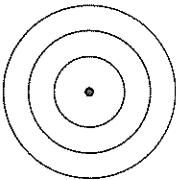
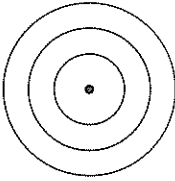
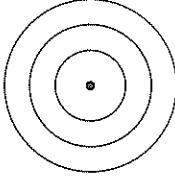
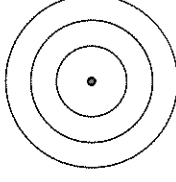
Questions

- 1 Complete the table for the elements indicated. The row for oxygen has been completed for you as an example. Use a periodic table to obtain the atomic number as required. A template has been provided for the electron shell models. Indicate the electrons using thick dots, as in the diagram above.

Element	Atomic number	Electron shell model	Electron configuration	Number of outer shell (valence) electrons
B				
N				

Figuring out electron configurations—page 2

Name: _____

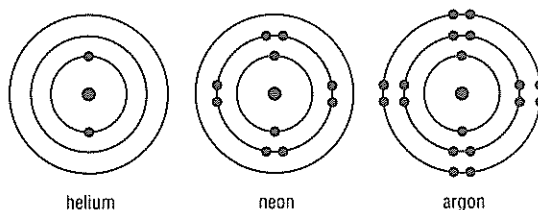
Element	Atomic number	Electron shell model	Electron configuration	Number of outer shell (valence) electrons
O	8		2 6	6
F				
Na				
Si				
P				

Figuring out electron configurations — page 3

Name: _____

- 2 Nitrogen and phosphorus are both group V elements. What is the relationship between the number of valence electrons and the group number?

- 3 Electron shell models for the first three noble gases are shown below.



Electron shell models for the first three noble gases

- a Write the electron configuration of these noble gases under the electron shell diagrams.
- b What is the relationship between the number of outer shell electrons and the electron configuration for the noble gases?
- _____
- _____
- c A characteristic of noble gases is that they are unreactive. Given that chemical reactions involve the sharing or transfer of electrons between atoms, explain why the noble gases may be so unreactive.

Bonding—it's all about electrons!—
page 1

Name: _____

Skill: applying

All chemical bonds involve electrons. Electrons can be transferred between atoms, as in **ionic bonding**, or shared between atoms, as in **covalent bonding**. What happens after the electrons are transferred or shared? In this activity you will explore the electron configurations of atoms after the formation of both ionic and covalent bonds.

Part A: Ionic bond formation and electron configuration

Figure 1 shows the formation of the ionic solid, sodium fluoride. The electron configuration (EC) for each of the atoms and the ions is also given.

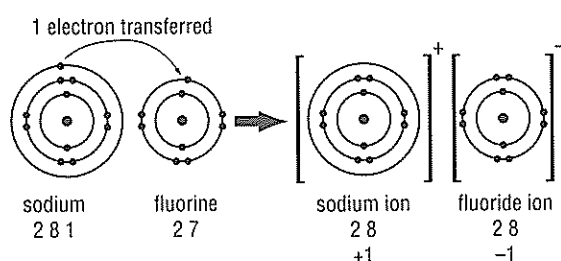


Fig 1 Formation of sodium fluoride

Questions

- 1 Why does the sodium ion have a +1 charge after the electron is transferred? Why does the fluorine atom form a -1 charge after the electron is transferred?

- 2 How many electrons are in the outer shell of both the sodium and the fluorine ions? What is so special about this number of electrons?

- 3 Figure 2 shows the formation of the ionic substance magnesium oxide. Complete the diagram by showing arrows representing the electron transfers, electron shell model arrangement of the ions, charge on the ions and finally the electron configuration of the ions formed.

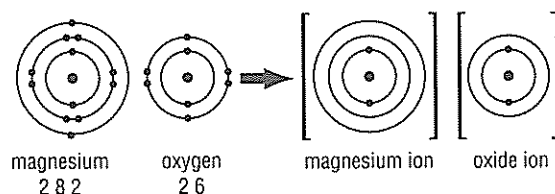


Fig 2 Formation of magnesium oxide

- 4 What is the number of electrons in the outer shell of both the magnesium and oxide ions, and how does this compare with your answer for question 2?

Bonding—it's all about electrons! — page 2

Extra for experts

- 5 Figure 3 shows the formation of the ionic substance aluminium sulfide (Al_2S_3). Complete the diagram by showing arrows representing the electron transfers, electron shell model arrangement of the ions, charge on the ions and finally the electron configuration of the ions formed.

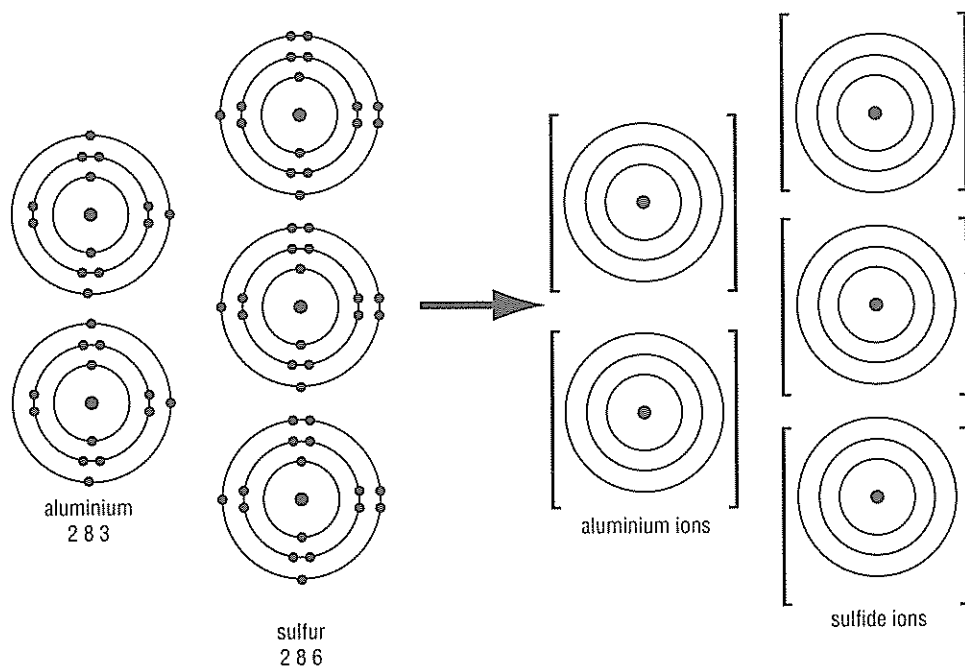


Fig 3 Formation of aluminum sulfide

Part B: Covalent bond formation and electron configuration

In covalent bonds, electrons are shared between atoms so that the atoms have noble gas electron configurations. Consider the formation of methane (CH_4). The diagrams show the electron configurations before sharing and after sharing. The diagram on the right replaces a single shared pair of electrons with a straight line.

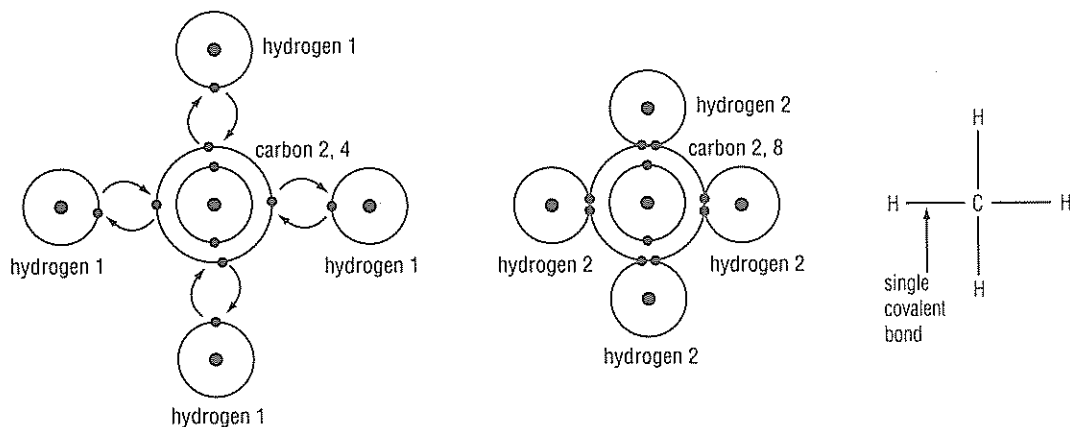


Fig 4 Formation of methane

Bonding—it's all about electrons!— page 3

Name: _____

Questions

- 6 What noble gas configuration do the carbon and the hydrogen have after covalent bond formation?
hydrogen = _____ carbon = _____
- 7 Figure 5 shows the formation of the covalent molecule carbon tetrachloride, CCl_4 . Complete the diagram by using arrows to show the sharing of the electrons, the electrons of the covalent molecule (the outer shell electrons have been removed from the carbon and the chlorine atoms) and then show how the molecule can be represented by replacing the covalent bonds with a line.

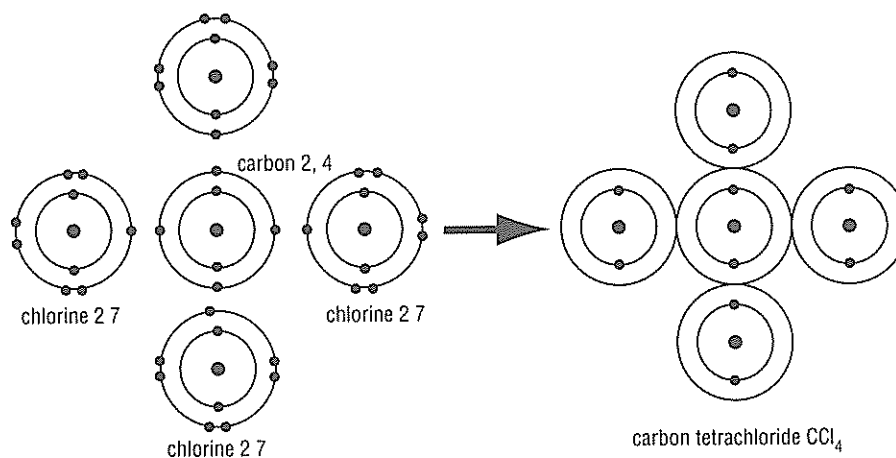


Fig 5 Formation of carbon tetrachloride

- 8 What noble gas configuration is obtained by the carbon and chlorine atoms after forming the covalent bond?
carbon = _____ chlorine = _____
- 9 In Figure 6 the electron dot diagram for one nitrogen (five electrons in outer shell) and three hydrogen atoms (one electron in outer shell) are shown. Draw one diagram that shows the formation of the ammonia molecule using arrows to represent the sharing of the electrons and a second diagram showing the electron dot diagram for the ammonia molecule.

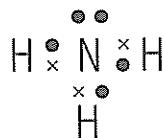


Fig 6 Formation of ammonia

Bonding—it's all about electrons! —
page 4

Name: _____

- 10 In the space below draw a diagram similar to that in question 9 showing the formation of carbon tetrachloride (CCl_4).

Extra for experts

When two pairs of electrons are shared between each atom, a double covalent bond is formed. The electron dot diagram in Figure 7 shows the double bond formation in the oxygen molecule.

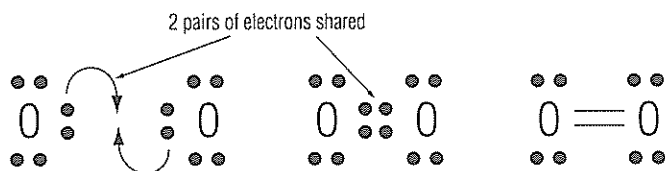


Fig 7 Formation of the double bond in oxygen

- 11 Draw a similar diagram to show the formation of carbon dioxide (CO_2).



Writing formulas of covalent molecules and ionic compounds—page 1

Name: _____

Skill: applying

It is very important to know the language of chemistry if you want to communicate with other scientists, from the person you sit next to in your science class to scientists in other countries. When you name compounds there are some guidelines that you need to follow. Chemistry is a science that has evolved and developed over many centuries. As a result there are a number of substances whose names have historical origins rather than conforming to a set of naming rules.

Naming covalent molecules

One of the largest groups of covalent compounds is organic compounds. You explored this group in *Science Aspects 3* and you will explore it further in your upper secondary science course.

When naming other covalent compounds you can follow these general guidelines:

- When two non-metals form a covalent molecule an 'ide' ending is used. The non-metal listed closer to the left-hand side of the periodic table is mentioned first.
- Terms such as 'mono' for one, 'di' for two and 'tri' for three are used to name compounds.
- If there is only one of the first named elements, the prefix 'mono' is not used.
- Diatomic elemental gases are just given their elemental name—for example, O_2 is oxygen gas, Cl_2 is chlorine gas.

1 Complete the table below by naming the following covalent molecular compounds:

Formula	Name of compound	Formula	Name of compound
CO		P_2O_5	
SO_2		N_2O_4	
SO_3		NI_3	
HCl		CF_4	
F_2		N_2	

2 Complete the table below by writing chemical formulas for the compounds listed.

Name of compound	Formula
Dihydrogen monoxide	
Oxygen difluoride	
Nitrogen trihydride (ammonia)	
Nitrogen trifluoride	
Nitrogen dioxide	

Writing formulas of covalent molecules and ionic compounds—page 2

Writing and naming ionic compounds

Use the guidelines and ion table below to help you complete the exercises that follow.

Writing formulas:

- Positive ions combine only with negative ions.
- Ions combine in such a way that the total charge is 0.
- When you are using polyatomic ions (radicals), if you use more than one, bracket the whole polyatomic ion and indicate the number of them with a subscript.

Naming ionic compounds:

- The overall ionic compound is not charged so the charge on the ions is not mentioned in the name of the compound. Prefixes are not used to indicate the number of each ion—just the name of the ion.
- The metal or, in the case of ammonium, the positive ion, is mentioned first.
- In the case of multivalent metals (such as iron, copper, tin and lead) the value of the ion being used is written in brackets following the name of the metal, for example iron (II) oxide, tin (IV) chloride.

Ion table

Positive ions	Symbol	Charge	Negative ions	Symbol	Charge
hydrogen	H ¹⁺	+1	fluoride	F ¹⁻	-1
sodium	Na ¹⁺	+1	chloride	Cl ¹⁻	-1
silver	Ag ¹⁺	+1	bromide	Br ¹⁻	-1
copper (I)	Cu ¹⁺	+1	iodide	I ¹⁻	-1
lithium	Li ¹⁺	+1	ethanoate	CH ₃ COO ¹⁻	-1
ammonium	NH ₄ ¹⁺	+1	hydrogen carbonate	HCO ₃ ¹⁻	-1
barium	Ba ²⁺	+2	hydroxide	OH ¹⁻	-1
calcium	Ca ²⁺	+2	nitrate	NO ₃ ¹⁻	-1
copper (II)	Cu ²⁺	+2	oxide	O ²⁻	-2
iron (II)	Fe ²⁺	+2	carbonate	CO ₃ ²⁻	-2
lead	Pb ²⁺	+2	sulfate	SO ₄ ²⁻	-2
magnesium	Mg ²⁺	+2	sulfide	S ²⁻	-2
tin (II)	Sn ²⁺	+2	phosphate	PO ₄ ³⁻	-3
zinc	Zn ²⁺	+2			
aluminium	Al ³⁺	+3			
chromium	Cr ³⁺	+3			
iron (III)	Fe ³⁺	+3			
tin (IV)	Sn ⁴⁺	+4			
lead (IV)	Pb ⁴⁺	+4			

You can also use the crossover method of writing the formulas for ionic compounds. This method was introduced in *Science Aspects 2*, page 257.

Writing formulas of covalent molecules and ionic compounds—page 3

Name: _____

- 3 Complete the table below by writing the names for the ionic compounds indicated:

Formula	Name	Formula	Name
Na_2S		$\text{Cu}(\text{NO}_3)_2$	
BaCl_2		$\text{Al}(\text{HCO}_3)_3$	
PbO		$\text{Zn}(\text{CH}_3\text{COO})_2$	
$\text{Zn}(\text{OH})_2$		Fe_2S_3	

- 4 Complete the table below by writing in the formula for the ionic compound formed by the intersection of the ions indicated. An example has been done for you.

Negative ions → Positive ions ↓	Bromide Br^{1-}	Ethanoate $\text{CH}_3\text{COO}^{1-}$	Nitrate NO_3^{1-}	Phosphate PO_4^{3-}	Hydrogen-carbonate HCO_3^{1-}
Aluminum Al^{3+}		$\text{Al}(\text{CH}_3\text{COO})_3$			
Copper (II) Cu^{2+}					
Barium Ba^{2+}					
Ammonium NH_4^{1+}					
Magnesium Mg^{2+}					

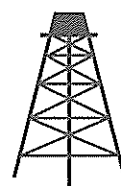
Recognising redox reactions



Name: _____

Skill: applying

Recognising oxidation and reduction equations can be difficult. A helpful mnemonic to use is 'OIL RIG'. Use this term to help you with the following problems.



OIL RIG
Oxidation Is Loss (of electrons)
Reduction Is Gain (of electrons)

Balancing half equations

1 Below are some incomplete half equations. Complete them by adding electrons to the correct side to balance the charge. Indicate whether each equation represents an oxidation or a reduction by crossing out the wrong term.

- | | | | |
|---|------------------|------------------------------|---------------------|
| a | Ag^+ | $\rightarrow \text{Ag}$ | oxidation/reduction |
| b | Na | $\rightarrow \text{Na}^+$ | oxidation/reduction |
| c | Zn | $\rightarrow \text{Zn}^{2+}$ | oxidation/reduction |
| d | Fe^{2+} | $\rightarrow \text{Fe}^{3+}$ | oxidation/reduction |
| e | Cu^{2+} | $\rightarrow \text{Cu}$ | oxidation/reduction |
| f | S | $\rightarrow \text{S}^{2-}$ | oxidation/reduction |
| g | Br_2 | $\rightarrow 2\text{Br}^-$ | oxidation/reduction |
| h | 2H^+ | $\rightarrow \text{H}_2$ | oxidation/reduction |
| i | 2O^{2-} | $\rightarrow \text{O}_2$ | oxidation/reduction |

Balancing redox reactions

To balance an overall redox reaction you need to remember that the electrons lost in the oxidation process must equal the number of electrons gained in the reduction process.

You can only balance the oxidation and reduction reactions by multiplying the entire half equation by the required coefficient.

2 The equations below are incomplete. For each pair of equations (one an oxidation and the other a reduction), balance the half equations (if required), indicate whether it is the oxidation or reduction process and then add the two half equations to give the redox reaction.

- | | | | |
|---|--|--|---------------------|
| a | Fe (s) | $\rightarrow \text{Fe}^{2+}$ | oxidation/reduction |
| | $\text{O}_2 \text{ (g)} + 4\text{e}^-$ | $\rightarrow \text{O}^{2-}$ | oxidation/reduction |
| | _____ | | redox/equation |
| b | Al (s) | $\rightarrow \text{Al}^{3+}$ | oxidation/reduction |
| | $\text{Cu}^{2+} \text{ (aq)}$ | $\rightarrow \text{Cu (s)}$ | oxidation/reduction |
| | _____ | | redox/equation |
| c | Na (s) | $\rightarrow \text{Na}^+ \text{ (aq)}$ | oxidation/reduction |
| | $\text{Cl}_2 \text{ (g)}$ | $\rightarrow \text{Cl}^- \text{ (aq)}$ | oxidation/reduction |
| | _____ | | redox/equation |



Name: _____

Skills: classifying, applying, analysing, evaluating

Kate and David were assigned the task of designing an investigation to determine their own metal activities series and compare it with the accepted metal activity series. For their investigation, they ordered the following substances:

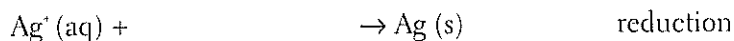
- Metals: small strips of magnesium, lead, iron, copper, zinc.
- Solutions: magnesium nitrate— $\text{Mg}(\text{NO}_3)_2$, lead nitrate— $\text{Pb}(\text{NO}_3)_2$, iron (II) nitrate— $\text{Fe}(\text{NO}_3)_2$, copper (II) nitrate— $\text{Cu}(\text{NO}_3)_2$, zinc nitrate— $\text{Zn}(\text{NO}_3)_2$.

They added the small strips of magnesium to each of the solutions and let them sit for a few minutes. They recorded whether or not crystals of metal appeared on the metal, showing that a displacement reaction had occurred.

They then repeated this investigation with all the other metals and recorded whether or not metal crystals appeared. A summary of their results are shown in the table below.

	Magnesium nitrate	Lead nitrate	Iron nitrate	Copper nitrate	Zinc nitrate
Magnesium nitrate	—	yes	yes	yes	yes
Lead nitrate	no	—	no	yes	no
Iron nitrate	no	yes	—	yes	no
Copper nitrate	no	no	no	—	no
Zinc nitrate	no	yes	yes	yes	—

- Given that a metal will displace any metal that is higher than it in the activity series from a solution of its salt, rank the metals above from most active to least active.
- Kate found a solution of silver nitrate but did not have any silver metal. She placed a strip of copper metal into the silver nitrate solution and observed that silver metal crystals were displaced from solution.
 - The teacher helped the two students by giving them the two incomplete half equations for this reaction. Complete the balancing of the half equations and then add them together to get the redox reaction.



- Based on this observation, where would you place silver metal in *your* metal activity series?

Name: _____

- 3 Kate and David were asked to write an evaluation of their investigation. They decided that an important part of the evaluation was to compare their results with known and accepted results. Kate located the following table that ranked the activity of the metals.

Accepted metal activity series	
	most reactive
Li	↓
K	
Ca	
Na	
Mg	
Al	
Mn	
Zn	
Cr	
Fe	
Sn	
Pb	
Cu	
Hg	
Ag	
Pt	
Au	
	least reactive

Write a sentence that describes their results compared with the accepted results.

- 4 Tin was not tested in the students' original investigation. They decide to test the tin by obtaining some tin strips and solutions of tin (II) nitrate. In the results table below, predict how you think the tin would have reacted.

	Magnesium nitrate	Lead nitrate	Iron nitrate	Copper nitrate	Zinc nitrate	Tin nitrate
Magnesium	–	yes	yes	yes	yes	
Lead	no	–	no	yes	no	
Iron	no	yes	–	yes	no	
Copper	no	no	no	–	no	
Zinc	no	yes	yes	yes	–	
Tin						

Metal activity series—page 3

Name: _____

Reactivity and the periodic table

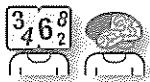
Kate and David wanted to see if the position of the metal in the periodic table had any relationship to the activity of the metal. They decided that the best way to do this would be to place their metals into a blank periodic table.

- 5 a Below is a blank periodic table. Place the metals from the accepted metal activity series into the table using a number ranking system. Number the most reactive metal '1' and place the number 1 in the corresponding position in the table. Use '2' for the next most reactive metal and place it in the table. Repeat until all of the metals have been placed.

I																		VIII
	II												III	IV	V	VI	VII	

← Transition metals →

- b In which group of the periodic table are the most active metals found? Explain why David and Kate would not have tested the activity of these metals in aqueous solution.
- _____
- _____
- c In which section of the periodic table are the least active metals found?
- _____



Name: _____

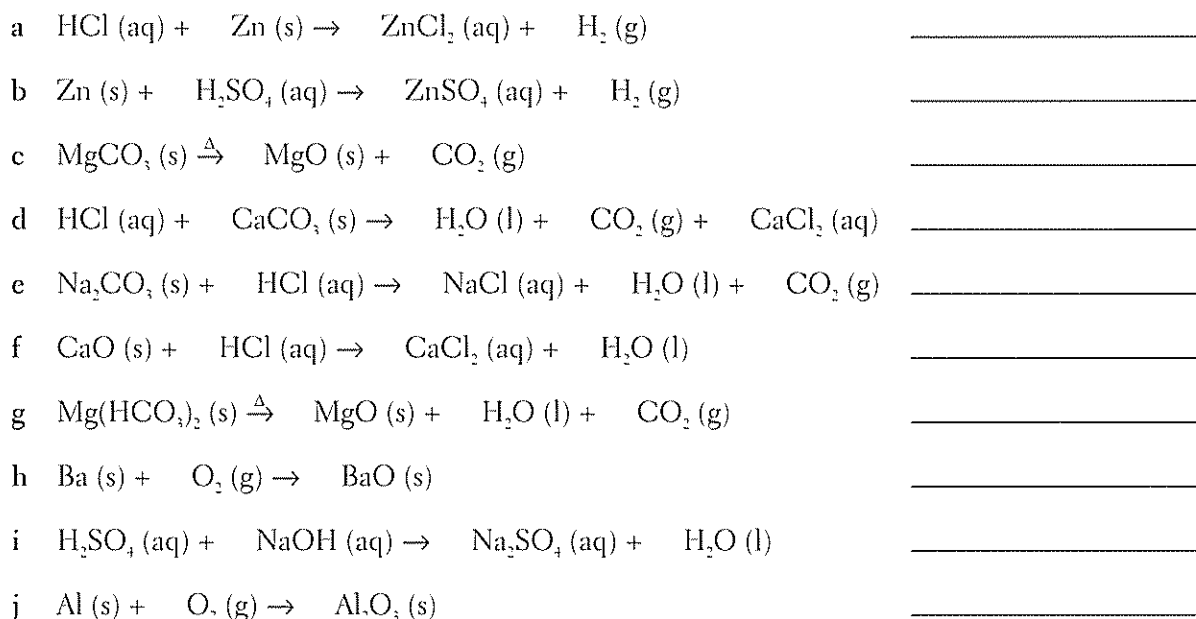
Skill: applying

Being able to predict whether or not a chemical reaction will occur and then being able to write a balanced chemical equation is an essential skill for a chemist.

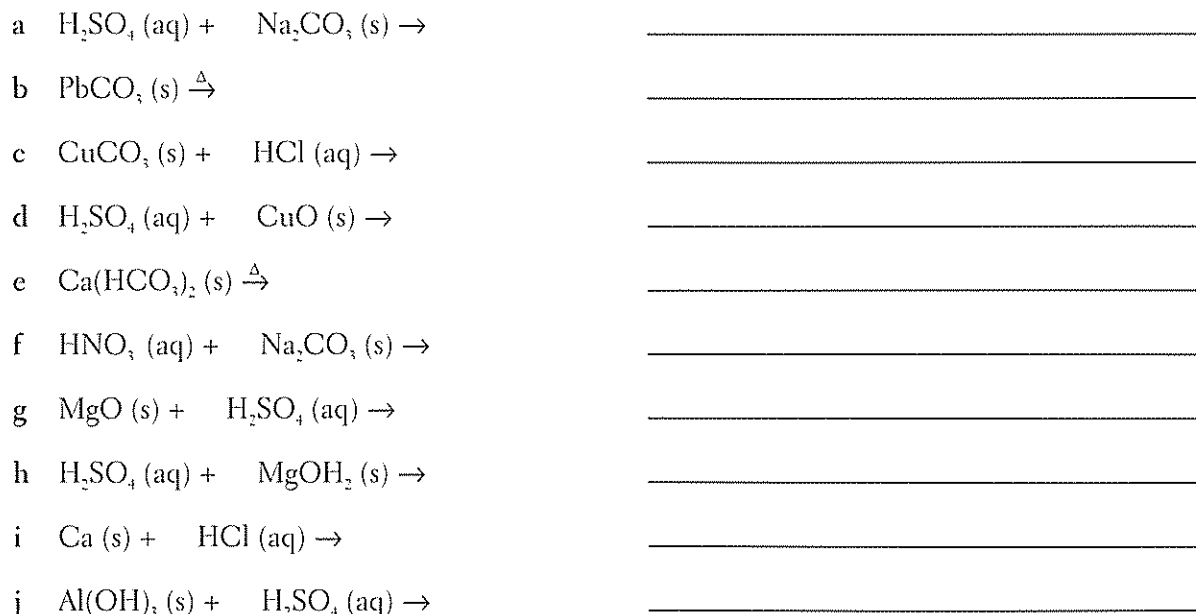
Write balanced chemical equations for the reactions shown below. Use your coursebook for the general equations, solubility rules and metal activity series as required.

Remember—you must show the phases of matter in all chemical equations. All equations are to be completed as molecular equations.

- 1 Balance the following equations. If an equation is already balanced, indicate by writing 'already balanced' next to it.



- 2 Complete and balance the following equations:



Name: _____

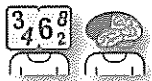
Precipitation reactions

- 3 Use the table of solubility rules to determine whether precipitation reactions will occur when the following aqueous solutions are mixed. Write balanced molecular equations and if a reaction does not occur, indicate 'no reaction'.
- a $\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow$ _____
- b $\text{Pb(NO}_3)_2 + \text{Na}_2\text{SO}_4 \text{ (aq)} \rightarrow$ _____
- c $\text{ZnCl}_2 \text{ (aq)} + \text{NaOH (aq)} \rightarrow$ _____
- d $\text{Al}_2\text{SO}_4 \text{ (aq)} + \text{MgCl}_2 \text{ (aq)} \rightarrow$ _____
- e $\text{H}_2\text{SO}_4 \text{ (aq)} + \text{BaCl}_2 \text{ (aq)} \rightarrow$ _____
- 4 Write balanced molecular equations for any reaction that occurs when the following solutions are mixed. If a reaction does not occur, write 'no reaction'.
- a sodium carbonate and copper (II) nitrate _____
- b aluminium chloride and ammonium hydroxide _____
- c potassium sulfate and lead nitrate _____
- d zinc chloride and silver nitrate _____
- e sodium iodide and lead nitrate. _____

Displacement reactions

- 5 Use the metal activity series to predict whether a reaction will occur under the situations described. Where a reaction does occur then write a balanced chemical equation. If a reaction does not occur, write 'no reaction'.
- a $\text{Zn (s)} + \text{Fe(NO}_3)_2 \text{ (aq)} \rightarrow$ _____
- b $\text{Al (s)} + \text{Ni(NO}_3)_2 \text{ (aq)} \rightarrow$ _____
- c $\text{Pb (s)} + \text{NaNO}_3 \text{ (aq)} \rightarrow$ _____
- d $\text{Cu (s)} + \text{Sn(NO}_3)_2 \text{ (aq)} \rightarrow$ _____
- e $\text{Fe (s)} + \text{CuSO}_4 \text{ (aq)} \rightarrow$ _____

Net ionic equations



Name: _____

Skill: applying

Write net ionic equations for the following reactions. Refer to your general equations, solubility rules and metal activity series from your coursebook as required. If no reaction occurs, indicate 'no reaction'.

- 1 Hydrochloric acid is added to solid sodium carbonate.

- 2 A solution of potassium hydroxide is added to a solution of zinc chloride.

- 3 A solution of magnesium sulfate is added to a solution of barium hydroxide.

- 4 A solution of ammonium sulfate is added to a solution of barium nitrate.

- 5 Sulfuric acid is added to aluminium oxide.

- 6 Zinc metal is added to a copper (II) sulfate solution.

- 7 Hydrochloric acid is added to magnesium metal.

- 8 Nitric acid is added to solid magnesium oxide.

- 9 Solutions of sodium carbonate and copper (II) nitrate are mixed.

- 10 Solutions of potassium sulfate and lead nitrate are mixed.
